

moonshotai/kimi-k2.7-code Routing, Cache, Cost, and Streaming Performance A/B Benchmark

Model: moonshotai/kimi-k2.7-code

This report compares Infron and OpenRouter using the standard prompt-cache-bench A/B methodology: 4 experiment groups, 50 rounds per group, streaming Chat Completions, short/medium/long prompt length tiers, and platform-default reasoning behavior.

Data Quality

- Completed round slots: 1600
- Request-level observations: 3200
- Paired samples: 800
- Excluded records: 0 total; 0 incomplete, 0 anomalous usage, 0 unequal input-token pairs.
- The aggregate tables below are computed from cleaned comparable records in `summary.json`.

Aggregate Results

Routing objective	Provider	Effective rounds	Cache hit rate	Total cost (USD)	Avg latency (ms)	P95 latency (ms)	Avg TTFT (ms)	P95 TTFT (ms)	Output TPS
throughput	infron	200	0.954646	0.980326	5,605.977	11,593.691	4,769.417	10,308.844	5.369
throughput	openrouter	200	0.976019	2.016535	2,759.595	4,844.239	2,593.428	4,541.029	5.153
price	infron	200	0.982922	0.944881	6,081.993	14,662.979	5,360.048	13,225.201	4.286
price	openrouter	200	0.929548	1.809493	10,768.469	37,562.774	9,952.846	29,846.345	1.368
latency	infron	200	0.916395	1.409992	8,638.226	16,980.644	7,891.779	15,814.239	2.407
latency	openrouter	200	0.974580	1.625742	7,154.683	15,814.692	6,888.809	14,972.450	2.001
ttft	infron	200	0.998297	0.941845	7,862.916	14,168.666	7,231.294	13,384.801	2.866
ttft	openrouter	200	0.997828	1.567765	6,955.335	14,086.271	6,741.560	13,847.978	2.084

Readout

- Infron recorded lower observed cost across all routing objectives in this run.
- Cache-hit leadership varied by routing objective; see `summary.json` and the HTML report for tier-level analysis and charts.

- Latency and TTFT leadership should be interpreted by routing objective because provider selection differs by `provider.sort`.
- No records were excluded after the standard data-quality and input-token pairing filters, so the run is suitable for direct paired comparison.

Reproducibility Links

- Summary: <<https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/data/summary.json>>
- Paired dataset CSV: <https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/data/benchmark_pairs.csv>
- Request-level dataset JSONL: <https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/data/benchmark_requests.jsonl>
- Filtered structured records: <<https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/data/records.json>>
- Excluded-record audit: <https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/data/records_excluded.json>
- Benchmark runner source: <https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/code/rerun_routing_sort_cache_cost_ab.py>
- HTML report renderer source: <https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/code/render_glm52_deepseek_style_report.py>
- Test source: <https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/code/test_rerun_routing_sort_cache_cost_ab.py>
- A/B report standard: <<https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/code/ab-report-standard.md>>
- Data directory: <<https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/data>>

- Code snapshot: <<https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/code>>
- English HTML report: <https://infronai.github.io/prompt-cache-bench/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/reports/routing-cache-cost-streaming-performance-ab-study__kimi-k2-7-code__infron-vs-openrouter__4x50-stream-ttft-reasoning-default-length-tiers-chat-completions__2026-07-10.en.html>
- Chinese HTML report: <https://infronai.github.io/prompt-cache-bench/experiments/moonshotai/kimi-k2.7-code/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-10/reports/routing-cache-cost-streaming-performance-ab-study__kimi-k2-7-code__infron-vs-openrouter__4x50-stream-ttft-reasoning-default-length-tiers-chat-completions__2026-07-10.zh.html>